

Assignment -1

Course : Database Management System (DBMS)

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① Ans:- A Database Management System (DBMS) is a software system that is used to store, manage & retrieve data in an organized way. It helps users to create database & perform operations like inserting, updating, deleting & searching data easily.

In simple words, DBMS acts as an interface b/w the user & the database.

→ Why DBMS is Required:

- 1) Reduces Data Redundancy :- It avoids duplication of data. Same data is not stored multiple times.
- 2) Maintains Data Consistency :- If data is updated in one place, it is updated everywhere.
- 3) Provides Data Security :- only authorized users can access or modify the data.
- 4) Data Integrity :- It ensures that the data stored is accurate & valid using constraints like primary key & foreign key.
- 5) Backup & Recovery :- DBMS provides backup facilities & can recover data in case of system failure.

6) Concurrency Control :- Multiple user can access the database at the same time without affecting each other.

② Data Abstraction :- Data Abstraction means hiding the internal details of how data is stored & showing only the necessary information to users.

→ It helps in reducing complexity & improving security.

→ Three levels of data abstraction :-

1) Physical level (Internal level) :-

→ lowest level.

→ Describes how data is actually stored in the database.

→ It includes storage details like files, indexes, memory, etc.

→ only database administrators (DBA) deal with this level.

2) Logical level (Conceptual level) :-

→ Describe what data is stored & relationship b/w data.

→ It includes tables, attributes & constraints.

→ Database designers work at this level.

3) View level (External level) :-

→ highest level

→ shows only required data to users.

→ Different users can have different views.

3
Supports Concurrency :- Multiple users can access the database at the same time without affecting each other's work.

5) Backup & Recovery :- If data is lost due to system failure, DBMS provides backup & recovery mechanisms.

6) Data Integrity :- It enforces rules like primary key & foreign key constraints to ensure accuracy of data.

3) Why is DBMS better than the traditional file system?

Ans:- Before DBMS was introduced, data was stored using traditional file systems. In this system, data was kept in separate files. & each application managed its own data independently. Although this method worked for small systems, it created many problems when the amount of data increased.

DBMS is considered better than the traditional file system for several important reasons.

1) Reduced data Redundancy :-

In a traditional file system, the same data is often stored in multiple files. For example, a student's details might be stored in admission records, exam records & library records separately. This leads to duplication of data.

2) Avoids Data Inconsistency :- When the same data is stored in multiple places, updating it in one file but not in another creates inconsistency. For example, if a student changes their phone number & it's updated in one file but not in another, the records become mismatched.

3) Better Data Security :- In a file system, there is very limited control over who can access or modify data. In DBMS, different users are given different access rights. For ex:-

- Admin can modify everything.
- Staff can update records.
- Students can only view their details.

u) Data Integrity:-

→ DBMS applies rules such as:

- Primary key (no duplicate values)
- Foreign key (maintains relationships b/w tables)
- NOT NULL, UNIQUE, CHECK constraints.

→ These rules ensure that data remains accurate & valid.

s) Concurrency Control:- In traditional systems, if two users try to access or modify the same file at the same time, conflicts may occur.

→ DBMS allows multiple users to work simultaneously without disturbing each other using concurrency control mechanisms.

e) Backup & Recovery:- File systems usually do not provide automatic recovery systems.

→ DBMS provides backup & recovery features, which help restore data in case of system failure or crash.

u) What is Data Abstraction? Explain its levels.

Ans:- Data abstraction is a concept in DBMS that means hiding the internal details of how data is stored & showing only the necessary information to the user. It helps in reducing complexity & makes the system easier to understand & use.

→ users do not need to know how data is physically stored in the computer. They only need to know how to access or use it.

Physical level (Internal level)

- This is the lowest level of abstraction.
- At this level, the database describes how the data is actually stored in memory. It includes details like:
 - Data storage structure
 - Indexing
 - Record format
 - Storage location
- These technical details are handled by database administrators (DBA), not by normal users.
- For ex:- users don't know whether data is stored in B-trees, files or blocks. They just retrieve it.

2) Logical level (Conceptual level) :-

- This is the middle level of abstraction.
- It describes what data is stored in the database & the relationships among the data.
- At this level, we define:
 - Tables
 - Attributes (columns)
 - Data types
 - Relationships
- For e.g. in a student database:
 - Student (Roll no, Name, Course)
 - Course (Course ID, Course Name)
- mainly used by database designers & developers.

3) View level (External level) :- → This is the highest level of abstraction.

- It shows only a specific part of the database according to the user's requirement.
- Different users can have different views of the same database.
- For example:
 - A student can see only their marks.
 - A teacher can see marks of all students.
 - An admin can see full records.
- This level improves security & simplicity.

→ why Data Abstraction is Important

- It reduces complexity
- It improves security
- It allows data independence
- It makes the system easier to manage & modify.

Ques: 5 Data Independence :-

Ans:- Data independence is the ability to change the structure of the database at one level without affecting the structure at another level. It is one of the most important advantages of DBMS.

→ It means we can make changes in the database design without disturbing users or application programs.

→ There are two types of data independence.

1) Physical Data Independence :- It refers to the ability to change the physical storage details without affecting the logical structure of the database.

→ For example:-

- changing the storage device (from HDD to SSD).
- Modifying indexing techniques.
- changing file organization.

→ These changes do not affect tables, attributes, or application programs.

→ This type of independence is easier to achieve.

2) Logical Data Independence :- It refers to the ability to change the logical structure without affecting the external views.

→ For example:

- Adding a new column to a table.
- Removing an attribute.
- Modifying relationships.

→ If a new attribute like "Email" is added to a student table, existing users who do not need it will not be affected.

→ Logical data independence is more difficult to achieve compared to physical data independence.

Why is Data Independence Needed?

- 1) It makes the database flexible.
- 2) It allows system upgrades without rewriting applications.
- 3) It reduces maintenance cost.
- 4) It improves long-term system stability.

7) Entity-Relationship (ER) Model.

Ans:- The ER Model is a conceptual model used in DBMS to represent real-world objects & the relationship b/w them in graphical form. It helps in designing the data structure before actually creating tables.

→ The ER model is like a blueprint of the database. It shows that what entities exist, what information they contain & how they are connected to each other.

Main Components of ER Model:-

1) Entity:- An entity is a real-world object or thing about which we store data.
Example:- Students, Teacher, Course, Book. (by rectangles)

2) Attributes:- are the properties or characteristics of an entity.
ex:- Student entity:- have Roll number, Name, Age, Course.

→ represented by ovals

3) Relationship:- shows how two or more entities are connected.

ex:- • Student enrolls in a Course • A Doctor treats a Patient.

→ represented by diamonds in ER diagrams.

→ Why important:-

- 1) It helps in clean database design before implementations.
- 2) It makes complex systems easier to understand.
- 3) It reduces errors in database structure.
- 4) It improves communication b/w developers & users.

8) What are Integrity Constraints? why are they important in DBMS?

Ans: Integrity constraints are rules applied to a database to maintain the accuracy, consistency, & reliability of data. These rules ensure that incorrect or invalid data cannot be entered into the database.

→ Integrity constraints protect the database from storing wrong information.

Types of Integrity Constraints :-

1) Primary key constraint :- A primary key uniquely identifies each record in a table.

- It cannot have duplicate values.

- It cannot be NULL.

2) Foreign key constraint :- A foreign key is used to maintain relationships b/w two tables.

→ It ensures that the value in one table must exist in another table.

→ This maintains referential integrity.

3) Unique constraints :- It ensures that all values in a column are different.
example: Email ID of students should be Unique.

4) Not NULL constraints :- It ensures that a column cannot have a NULL (empty) value.

→ Ex: - Name field should not be left empty.

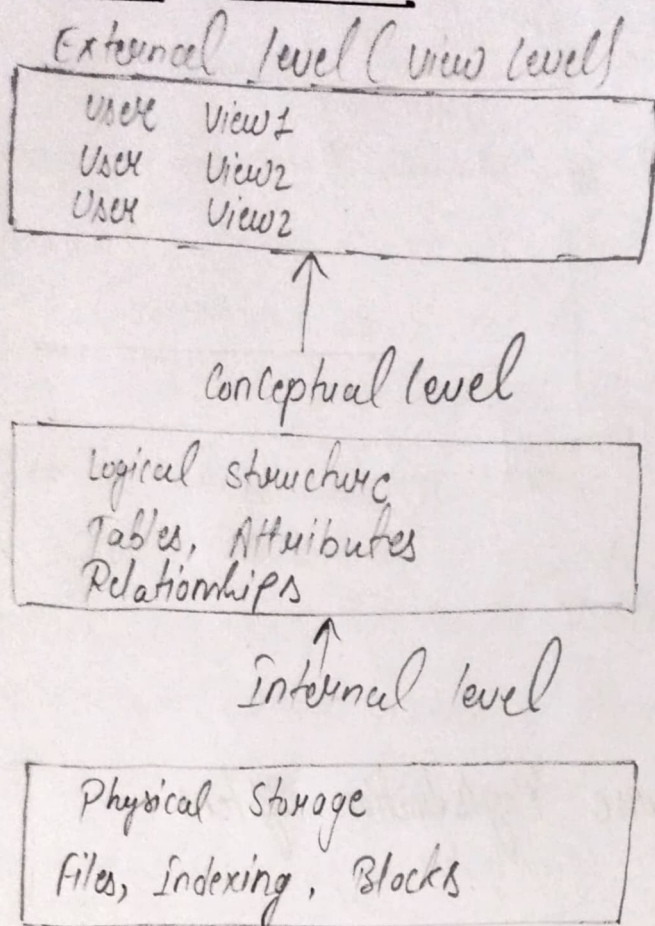
5) check constraint :- It ensures that the value entered satisfies a specific condition.

ex: - $\text{age} \geq 18$

$\rightarrow 0 \leq \text{marks} \leq 100$

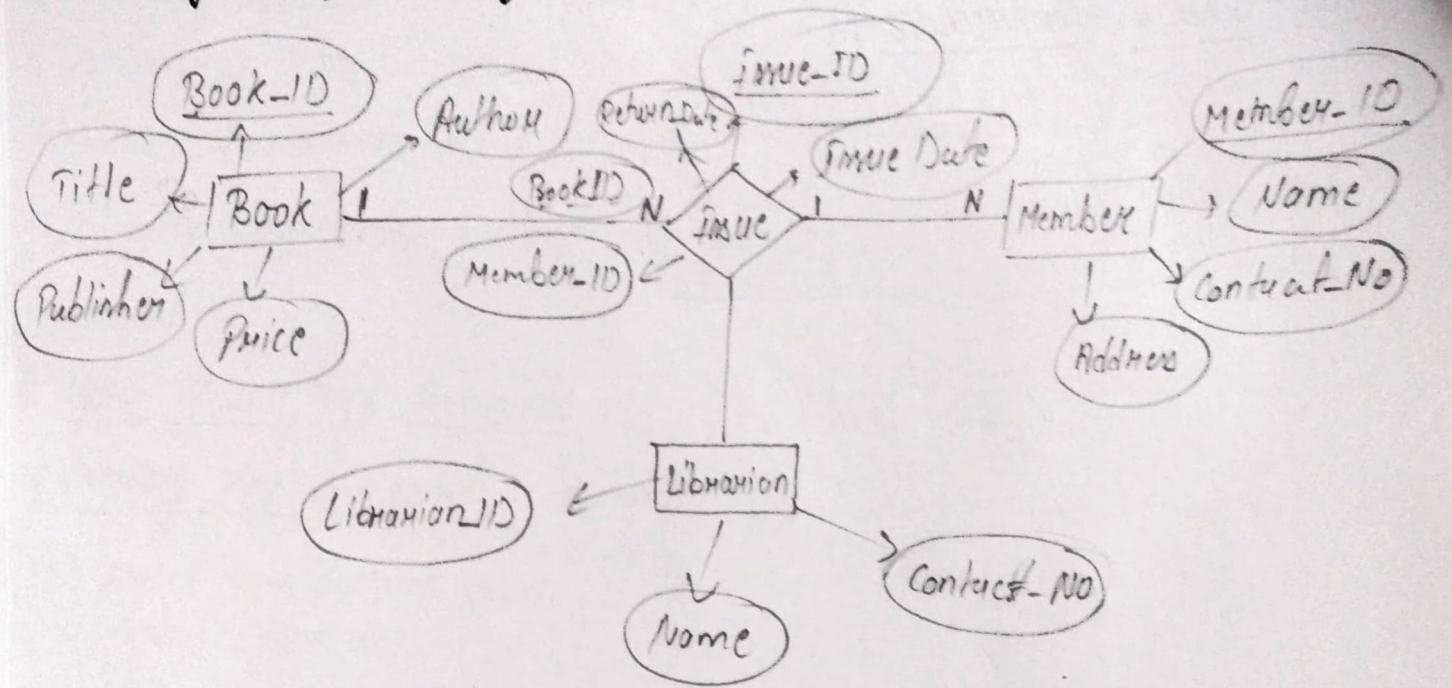
9)

Three-Schema Architecture :-

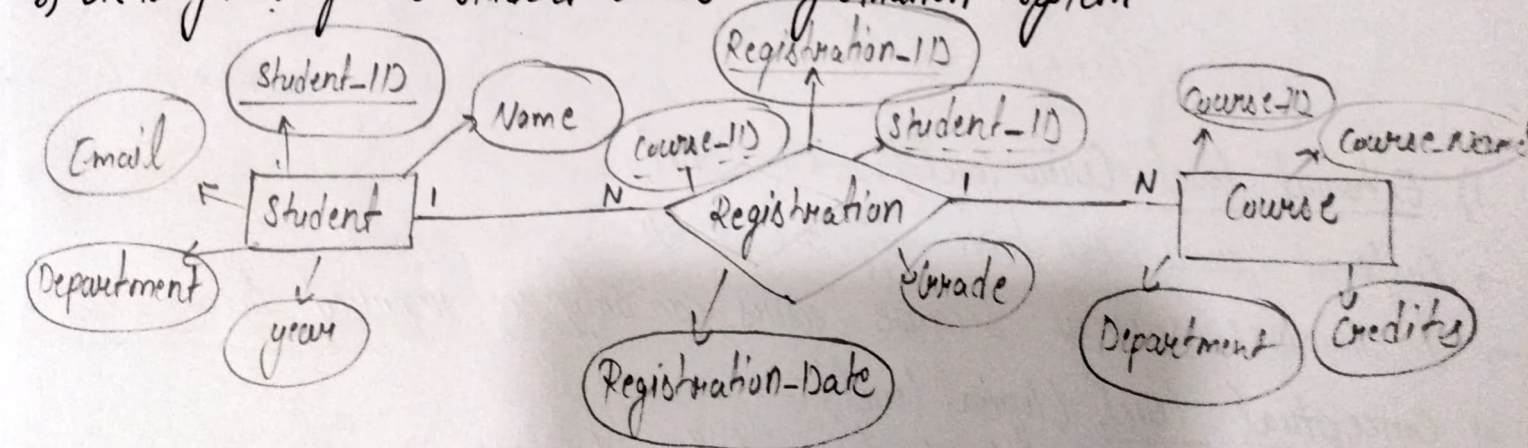


- 1) External level (View level) :- → Top level.
 - Different users can have different views.
 - It improves security because users can only see required data.
- 2) Conceptual level (logical level) :-
 - It describes the overall structure of the database.
 - It includes tables, attributes, relationship & constraints.
 - It does not include physical storage details.
- 3) Internal level (Physical level) :-
 - It explains how data is physically stored
 - Including indexing, file organization, storage structure.
 - The biggest benefit is Data Independence.

1) Library Management Systems



2) ER Diagram for a Student-Course Registration System



Section-C

1) Entities

1) Patient

2) Doctor

3) Appointment

→ Attributes

1) Patient

→ Patient-ID (PK)

→ Name

→ Age

→ Gender

→ Contact-No

2) Doctor

→ Doctor-ID (PK)

→ Name

→ Specialization

→ Contact-No

3) Appointment

→ Appointment-ID (PK)

→ Date

→ Time

→ Patient-ID (FK)

→ Doctor-ID (FK)

→ Relationships

→ Patient book an Appointment

→ A Doctor attends an Appointment

→ Cardinality

→ Patient → Appointment (1:N)

→ Doctor → Appointment (1:N)

Entities:-

- 1) Customer → Primary keys → Customer-ID
- 2) Product → Product-ID
- 3) Order → Order-ID

→ Foreign keys

- Customer-ID in order Table (references Customer)
- Product-ID in order-Item table (if created)

→ Constraints:-

- Email should be Unique.
- Price should be greater than 0 (Check constraint).
- Stock should not be -ve.
- order must reference an existing Customer.

2) A University stores student records. Explain how data abstraction helps different users.

Ans:- Students can view their marks & attendance.

- Teachers can update marks of their subjects.
- Admin can access full student details.

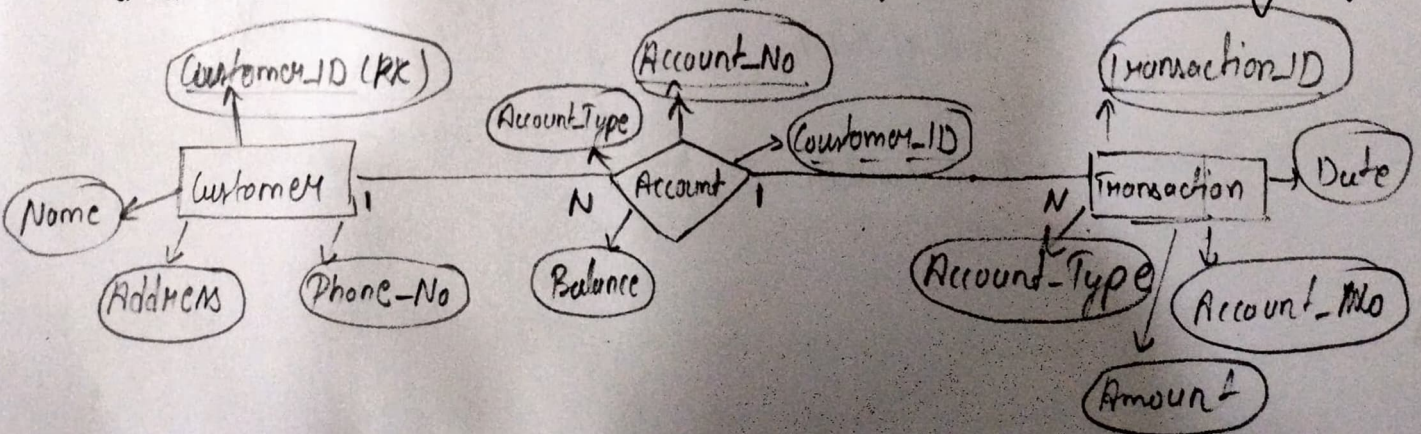
→ Three level abstraction:-

- view level - different users see different views
- Logical level - Defines table like student, Course, Marks.
- Physical level - Stores data internally without users knowing details.

→ data abstraction improves security, reduces complexity & allows smooth functioning.

Section - D

1) Draw a simple ER diagram for any-life system & explain one key design decision.



→ one key Design Decision (Insight)

→ created a separate Transaction entity instead of storing transaction details inside the account entity.

→ Reasons :- 1 accounts can have many transactions

→ Avoids storing repeated transaction data in the Account table

→ makes the system more organized & scalable

→ It improves data consistency.

Section D

1) Three-Schema Architecture :-

→ Benefit :-

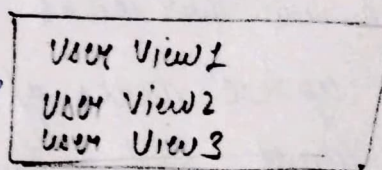
→ The main benefit is Data Independence

→ This means changes at one level do not affect other levels.

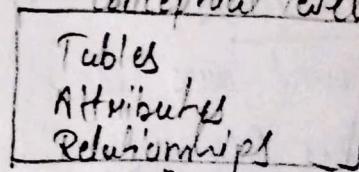
Ex :-

→ If storage structure is changed at the internal level, user at the external level will not be affected.

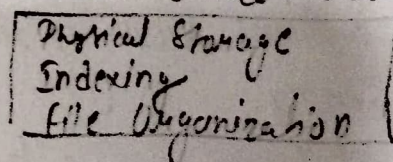
External level (View level)



Conceptual level



Internal level



2) Student Entity :- → Attributes :-

- Student-ID (PK)
- Name
- Date-of-Birth
- Email
- Phone-Number
- Course

